C

ACT SUBJECT TEST — MATH 2 • UNIT 4.01

Advanced Functions, Polynomials & Logarithms

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 A	Q3 A	Q4 A	Q5 D
Q6 C	Q7 A	Q8 A	Q9 B	Q10 B
Q11 A	Q12 D	Q13 A	Q14 B	Q15 D
Q16 B	Q17 B	Q18 A	Q19 B	Q20 D
Q21 C	Q22 C	Q23 C	Q24 C	Q25 B

Advanced Functions, Polynomials & Logarithms Practice Exercises

1 MCQ

The graph of the function $f(x) = (x-1)^2 - 4$ is shown in the standard (x, y) coordinate plane below. What is the range of f?

A $[-4, \infty)$ ✓

B $[1, \infty)$

C $(-\infty, -4]$

D $(-\infty, \infty)$

WORKED SOLUTION & VERIFICATION

1. **Identify the vertex and direction of opening**: The function $f(x) = (x-1)^2 - 4$ is written in vertex form, $f(x) = a(x-h)^2 + k$, where $a = 1$, $h = 1$, and $k = -4$. The vertex of the parabola is at $(1, -4)$.
2. **Determine the minimum output value**: Since the leading coefficient $a = 1 > 0$, the parabola opens upward, meaning that the vertex represents the absolute minimum point on the graph. The minimum value attained by $f(x)$ is -4.
3. **Determine the range**: The range represents all possible output values (y-values) of the function. Because the function takes on its minimum value of -4 at $x = 1$ and increases without bound as x moves away from 1, the range is all real numbers greater than or equal to -4, which is written in interval notation as $[-4, \infty)$.

Common Misconceptions:

- Choice B, $[1, \infty)$, incorrectly uses the x-coordinate of the vertex instead of the y-coordinate.
- Choice C, $(-\infty, -4]$, incorrectly assumes the parabola opens downward.
- Choice D, $(-\infty, \infty)$, confuses the range (the set of y-values) with the domain (the set of all real x-values).

2 MCQ

The polynomial function $P(x) = 2x^3 + ax^2 - 5x + b$ has $(x - 1)$ as a factor. When $P(x)$ is divided by $(x + 2)$, the remainder is -18. What is the remainder when $P(x)$ is divided by $(x - 3)$?

- **A** 2 ✓
- **B** 8
- **C** 18
- **D** -30

WORKED SOLUTION & VERIFICATION

According to the Factor Theorem and the Remainder Theorem, if a polynomial $P(x)$ is divided by $(x - c)$, the remainder is $P(c)$. If $(x - c)$ is a factor, then $P(c) = 0$.

Step 1: Set up the first equation using the factor $(x - 1)$:

$$P(1) = 2(1)^3 + a(1)^2 - 5(1) + b = 0$$

$$2 + a - 5 + b = 0 \implies a + b = 3$$

Step 2: Set up the second equation using the divisor $(x + 2) = (x - (-2))$ with remainder -18:

$$P(-2) = 2(-2)^3 + a(-2)^2 - 5(-2) + b = -18$$

$$2(-8) + 4a + 10 + b = -18$$

$$-16 + 10 + 4a + b = -18 \implies 4a + b = -12$$

Step 3: Solve the system of linear equations for a and b:

Subtracting the first equation from the second:

$$(4a + b) - (a + b) = -12 - 3$$

$$3a = -15 \implies a = -5$$

Substitute $a = -5$ into $a + b = 3$:

$$-5 + b = 3 \implies b = 8$$

Thus, the polynomial is $P(x) = 2x^3 - 5x^2 - 5x + 8$.

Step 4: Find the remainder when $P(x)$ is divided by $(x - 3)$ by evaluating $P(3)$:

$$P(3) = 2(3)^3 - 5(3)^2 - 5(3) + 8 = 2(27) - 5(9) - 15 + 8 = 54 - 45 - 15 + 8 = 2$$

Common Pitfalls:

- Evaluating $P(2)$ instead of $P(-2)$ produces $4a + b = -24$, leading to $a = -9$, $b = 12$, and an incorrect remainder of -30.
- Confusing the constant term $b = 8$ with the final answer yields 8.

3 MCQ

A function f is defined for all real numbers except $x = 0$ and $x = 1$. For all values of x in its domain, f satisfies the functional equation

$$f(x) + f\left(\frac{x-1}{x}\right) = 1 + x.$$

What is the value of $f(2)$?

- **A** $\frac{3}{4}$ ✓
- **B** $\frac{5}{4}$
- **C** $\frac{7}{4}$
- **D** $\frac{9}{4}$

WORKED SOLUTION & VERIFICATION

To find $f(2)$, substitute strategic values of x to generate a system of equations involving the cyclic transformation $g(x) = \frac{x-1}{x}$.

1. Substitute $x = 2$:

$$\frac{2-1}{2} = \frac{1}{2}$$

$$f(2) + f\left(\frac{1}{2}\right) = 1 + 2 = 3 \quad \text{— (Equation 1)}$$

2. Substitute $x = \frac{1}{2}$:

$$\frac{\frac{1}{2}-1}{\frac{1}{2}} = \frac{-\frac{1}{2}}{\frac{1}{2}} = -1$$

$$f\left(\frac{1}{2}\right) + f(-1) = 1 + \frac{1}{2} = \frac{3}{2} \quad \text{— (Equation 2)}$$

3. Substitute $x = -1$:

$$\frac{-1-1}{-1} = \frac{-2}{-1} = 2$$

$$f(-1) + f(2) = 1 + (-1) = 0 \quad \text{— (Equation 3)}$$

Now solve this system of 3 linear equations for $f(2)$:

From Equation 3, we have $f(-1) = -f(2)$. Substituting this into Equation 2 gives:

$$f\left(\frac{1}{2}\right) - f(2) = \frac{3}{2} \quad \text{— (Equation 4)}$$

Subtract Equation 4 from Equation 1:

$$\left[f(2) + f\left(\frac{1}{2}\right)\right] - \left[f\left(\frac{1}{2}\right) - f(2)\right] = 3 - \frac{3}{2}$$

$$2f(2) = \frac{3}{2} \implies f(2) = \frac{3}{4}.$$

Common misconception: Solving for $f\left(\frac{1}{2}\right)$ instead yields $\frac{9}{4}$, which corresponds to a common distractor.

4 MCQ

Given the functions $f(x) = 3x - 4$ and $g(x) = x^2 + 5$, what is the value of $f(g(2))$?

- **A** 23 ✓
- **B** 9
- **C** 18
- **D** 27

WORKED SOLUTION & VERIFICATION

To find the value of the composite function $f(g(2))$, evaluate the inner function first, and then substitute that result into the outer function:

1. Evaluate $g(2)$:

$$g(2) = (2)^2 + 5 = 4 + 5 = 9$$

2. Evaluate $f(9)$:

$$f(9) = 3(9) - 4 = 27 - 4 = 23$$

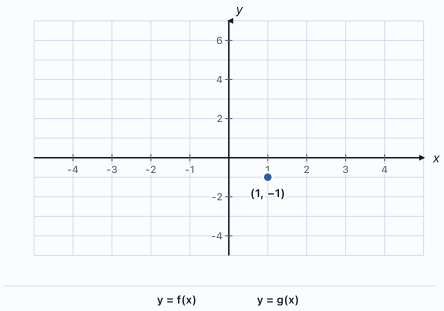
Common Misconceptions:

- Evaluating in the reverse order, $g(f(2))$, yields $f(2) = 3(2) - 4 = 2$, and $g(2) = 2^2 + 5 = 9$.
- Interpreting the expression as the product $f(2) \cdot g(2)$ yields $2 \cdot 9 = 18$.
- Evaluating $3 \cdot g(2)$ but omitting the constant -4 yields 27.

5 MCQ

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown in the standard (x, y) coordinate plane below, with vertex $(2, -1)$ and y-intercept $(0, 3)$.

The function g is defined by $g(x) = f(x + k) - 8$, where k is a constant. If the graph of g passes through the origin $(0, 0)$, which of the following could be the value of k?



A -5

B -3

C 1

D 5 ✓

WORKED SOLUTION & VERIFICATION

Step 1: Find the equation of $f(x)$.

Using the vertex form of a parabola with vertex $(h, k_v) = (2, -1)$:

$$f(x) = a(x - 2)^2 - 1$$

Since the graph passes through the y-intercept $(0, 3)$, substitute $x = 0$ and $y = 3$:

$$3 = a(0 - 2)^2 - 1 \implies 3 = 4a - 1 \implies 4a = 4 \implies a = 1$$

Thus, $f(x) = (x - 2)^2 - 1$.

Step 2: Apply the condition that g passes through $(0, 0)$.

Since $g(0) = 0$ and $g(x) = f(x + k) - 8$:

$$g(0) = f(0 + k) - 8 = f(k) - 8 = 0 \implies f(k) = 8$$

Step 3: Solve for k.

Substitute k into f:

$$(k - 2)^2 - 1 = 8$$

$$(k - 2)^2 = 9$$

$$k - 2 = 3 \quad \text{or} \quad k - 2 = -3$$

$$k = 5 \quad \text{or} \quad k = -1$$

Among the given choices, 5 is the correct value.

Common Algebraic Traps:

- Choice A (-5) arises from a sign error when writing the vertex form as $a(x + 2)^2 - 1$, yielding $(k + 2)^2 = 9 \implies k = -2 - 3 = -5$.
- Choice B (-3) arises from omitting the horizontal shift of the vertex and setting $k = -3$ directly from ± 3.
- Choice C (1) arises from solving $k + 2 = 3$, or confusing k with an x-intercept of $f(x)$.

6 MCQ

What is the minimum value of the function

$$f(x) = \sqrt{x^2 - 4x + 13} + \sqrt{x^2 - 14x + 130}$$

for all real numbers x?

A 12

B $4\sqrt{10}$

C 13 ✓

D $\sqrt{170}$

WORKED SOLUTION & VERIFICATION

To find the minimum value of $f(x)$, synthesize the algebraic expressions with coordinate geometry by completing the square under each radical:

1. Rewrite the radicands:

$$x^2 - 4x + 13 = (x-2)^2 + 9 = (x-2)^2 + 3^2$$

$$x^2 - 14x + 130 = (x-7)^2 + 81 = (x-7)^2 + 9^2$$

2. Interpret geometrically using the Euclidean distance formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$:

- $\sqrt{(x-2)^2 + 3^2}$ is the distance from the point $P(x, 0)$ on the x-axis to the fixed point $A(2, 3)$.
- $\sqrt{(x-7)^2 + 9^2}$ is the distance from $P(x, 0)$ to the fixed point $B(7, 9)$.

Thus, $f(x) = PA + PB$. We are looking for the point $P(x, 0)$ that minimizes the sum of distances $PA + PB$.

3. Apply the reflection principle and the Triangle Inequality:

Reflect point $A(2, 3)$ across the x-axis to obtain $A'(2, -3)$. For any point P on the x-axis, $PA = PA'$. Therefore:

$$PA + PB = PA' + PB \geq A'B$$

with equality occurring if and only if P lies on the straight line segment connecting $A'(2, -3)$ and $B(7, 9)$.

4. Calculate the length of segment $A'B$:

$$A'B = \sqrt{(7-2)^2 + (9-(-3))^2} = \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$$

The line segment connecting $A'(2, -3)$ and $B(7, 9)$ intersects the x-axis at $x = \frac{13}{4} = 3.25$, which is a valid real value of x. Hence, the minimum value is exactly 13.

Common Misconceptions:

- **Choice A (12):** Adding the individual minimums of the two radical expressions: $\min \sqrt{(x-2)^2 + 9} = 3$ (at $x = 2$) and $\min \sqrt{(x-7)^2 + 81} = 9$ (at $x = 7$). Because these minima occur at different values of x, they cannot be attained simultaneously.
- **Choice B ($4\sqrt{10}$):** Miscalculating the horizontal distance as $\Delta x = 7 - 3 = 4$ rather than $7 - 2 = 5$, leading to $\sqrt{4^2 + 12^2} = \sqrt{160} = 4\sqrt{10}$.
- **Choice D ($\sqrt{170}$):** Erroneously combining the two radicands directly under a single square root.

7 MCQ

The function f is defined by $f(x) = \frac{3x+2}{x-4}$ for all real numbers $x \neq 4$. What is the value of $f^{-1}(2)$?

A -10 ✓

B -6

C -4

D 6

WORKED SOLUTION & VERIFICATION

By definition of the inverse function, finding $k = f^{-1}(2)$ is equivalent to solving $f(k) = 2$.

Set up the equation:

$$\frac{3k+2}{k-4} = 2$$

Multiply both sides by $k-4$ (where $k \neq 4$):

$$3k + 2 = 2(k-4)$$

$$3k + 2 = 2k - 8$$

Subtract $2k$ from both sides:

$$k + 2 = -8$$

Subtract 2 from both sides:

$$k = -10$$

Thus, $f^{-1}(2) = -10$.

Common misconceptions:

- Evaluating $f(2) = \frac{3(2)+2}{2-4} = \frac{8}{-2} = -4$ confuses the function f with its inverse f^{-1}.
- A sign error when solving $3k + 2 = 2k - 8$ can lead to $k = -6$ or $k = 6$.

8 MCQ

The polynomial function $P(x) = x^3 + ax^2 + bx + 12$, where a and b are real constants, has $x - 1$ as a factor. When $P(x)$ is divided by $x - 2$, the remainder is -10. What is the greatest root of the equation $P(x) = 0$?

- **A** 4 ✓
- **B** 3
- **C** 1
- **D** 6

WORKED SOLUTION & VERIFICATION

To find the roots of $P(x) = 0$, we first determine the unknown parameters a and b using the Factor Theorem and the Remainder Theorem.

1. **Apply the Factor Theorem:**

Since $x - 1$ is a factor of $P(x)$, we have $P(1) = 0$:

$$P(1) = (1)^3 + a(1)^2 + b(1) + 12 = 0 \implies 1 + a + b + 12 = 0 \implies a + b = -13$$

2. **Apply the Remainder Theorem:**

When $P(x)$ is divided by $x - 2$, the remainder is -10, so $P(2) = -10$:

$$P(2) = (2)^3 + a(2)^2 + b(2) + 12 = -10 \implies 8 + 4a + 2b + 12 = -10 \implies 4a + 2b = -30$$

Dividing by 2 gives:

$$2a + b = -15$$

3. **Solve the linear system:**

Subtract the first equation from the second:

$$(2a + b) - (a + b) = -15 - (-13) \implies a = -2$$

Substitute $a = -2$ back into $a + b = -13$:

$$-2 + b = -13 \implies b = -11$$

Thus, the polynomial is:

$$P(x) = x^3 - 2x^2 - 11x + 12$$

4. **Factor $P(x)$ and find the roots:**

Since $x - 1$ is a factor, divide $P(x)$ by $x - 1$:

$$x^3 - 2x^2 - 11x + 12 = (x - 1)(x^2 - x - 12)$$

Factoring the quadratic factor gives:

$$x^2 - x - 12 = (x - 4)(x + 3)$$

So the full factorization is:

$$P(x) = (x - 1)(x - 4)(x + 3)$$

The roots of $P(x) = 0$ are $x = 1$, $x = 4$, and $x = -3$.

The greatest root is 4.

Common Traps:

- Factoring $x^2 - x - 12$ incorrectly as $(x + 4)(x - 3)$ leads to roots $-4, 1, 3$, with 3 mistakenly identified as the greatest root.
- Only noting the given root from the factor $x - 1$ gives 1.

9 MCQ

A line L is tangent to the graph of the polynomial function $f(x) = x^4 - 6x^2 + 5x + 7$ at two distinct points in the xy-plane. What is the y-intercept of line L?

A 2

B -2 ✓

C 7

D 16

WORKED SOLUTION & VERIFICATION

Let the equation of the tangent line L be $y = mx + b$.

Because L is tangent to the quartic polynomial $f(x)$ at two distinct points, say $x = r$ and $x = s$ with $r \neq s$, each point of tangency corresponds to a root of multiplicity at least 2 for the difference equation $f(x) - (mx + b) = 0$.

Since $f(x) - (mx + b)$ is a monic polynomial of degree 4, it factors completely over the real numbers as:

$$f(x) - (mx + b) = (x - r)^2(x - s)^2$$

Expanding this product:

$$(x - r)^2(x - s)^2 = (x^2 - (r + s)x + rs)^2$$

$$= x^4 - 2(r + s)x^3 + ((r + s)^2 + 2rs)\,x^2 - 2rs(r + s)x + (rs)^2$$

Now consider the left-hand side from the given function:

$$f(x) - (mx + b) = x^4 - 6x^2 + (5 - m)x + (7 - b)$$

Equating coefficients of corresponding powers of x:

1. **Coefficient of x^3:**

$$-2(r + s) = 0 \implies r + s = 0 \implies s = -r$$

Because $r \neq s$, we must have $r \neq 0$.

2. **Coefficient of x^2:**

Since $r + s = 0$, the x^2 coefficient simplifies to $2rs = 2r(-r) = -2r^2$:

$$-2r^2 = -6 \implies r^2 = 3$$

This gives two distinct real points of tangency at $x = \pm\sqrt{3}$.

3. **Coefficient of x:**

$$-2rs(r + s) = 0 \implies 5 - m = 0 \implies m = 5$$

4. **Constant term:**

$$(rs)^2 = (-r^2)^2 = (-3)^2 = 9$$

Therefore:

$$7 - b = 9 \implies b = 7 - 9 = -2$$

Thus, the equation of the bitangent line is $y = 5x - 2$, and its y-intercept is -2.

Common Pitfalls:

- Selecting 2 results from solving $7 - b = 9$ incorrectly as $b = 9 - 7$.
- Selecting 7 confuses the y-intercept of the line L with the y-intercept of $f(x)$, which is $f(0) = 7$.
- Selecting 16 results from a sign error when isolating b, computing $b = 7 + 9$.

10 MCQ

In the standard (x, y) coordinate plane, the graph of $y = f(x)$ contains the point $(-2, 5)$. The function g is defined by $g(x) = f(x - 3) - 4$. Which of the following points must lie on the graph of $y = g(x)$?

A $(-5, 1)$

B $(1, 1)$ ✓

C $(1, 9)$

D $(-5, 9)$

WORKED SOLUTION & VERIFICATION

We are given that the point $(-2, 5)$ lies on the graph of $y = f(x)$, which means $f(-2) = 5$.
The function g is defined as $g(x) = f(x - 3) - 4$.
To evaluate $g(x)$ using our known value $f(-2)$, we set the argument of f equal to -2:

$$x - 3 = -2 \implies x = 1.$$

Now, substitute $x = 1$ into the expression for $g(x)$:

$$g(1) = f(1 - 3) - 4 = f(-2) - 4 = 5 - 4 = 1.$$

Therefore, the point $(1, 1)$ must lie on the graph of $y = g(x)$.
Geometrically, the transformation $y = f(x - 3) - 4$ translates the graph of $y = f(x)$ by 3 units to the right and 4 units downward:

$$(x, y) \to (x + 3, y - 4).$$

Applying this translation to $(-2, 5)$ gives:

$$(-2 + 3, 5 - 4) = (1, 1).$$

Common Misconceptions:
- Choice B, $(-5, 1)$, results from incorrectly shifting left by 3 units instead of right.
- Choice C, $(1, 9)$, results from incorrectly shifting upward by 4 units instead of downward.
- Choice D, $(-5, 9)$, applies both incorrect directional shifts.

11 MCQ

If $f(x) = 5x + 4$ and $g(x) = -4x$, find $f(g(2))$.

A -36 ✓

B 6

C -56

D -31

WORKED SOLUTION & VERIFICATION

Applying Advanced Functions & Equations principles, $g(2) = -8$, then $f(-8) = -36$. Therefore, the verified correct option is -36.

12 MCQ

If $f(x) = 2x - 7$ and $g(x) = 4x - 1$, find $f(g(-2))$.

A -19	B -45
C -23	D **-25** ✓

WORKED SOLUTION & VERIFICATION

Applying Advanced Functions & Equations principles, $g(-2) = -9$, then $f(-9) = -25$. Therefore, the verified correct option is -25.

13 MCQ

If $f(x) = -2x + 7$ and $g(x) = 4x - 3$, find $f(g(1))$.

A **5** ✓	B 9
C 17	D 3

WORKED SOLUTION & VERIFICATION

Applying Advanced Functions & Equations principles, $g(1) = 1$, then $f(1) = 5$. Therefore, the verified correct option is 5.

14 MCQ

If $f(x) = -3x + 5$ and $g(x) = 3x + 5$, find $f(g(5))$.

A 5	B **-55** ✓
C -25	D -58

WORKED SOLUTION & VERIFICATION

Applying Advanced Functions & Equations principles, $g(5) = 20$, then $f(20) = -55$. Therefore, the verified correct option is -55.

15 MCQ

If $f(x) = -5x + 8$ and $g(x) = -x + 6$, find $f(g(-1))$.

A 14	B -7
C -32	D **-27** ✓

WORKED SOLUTION & VERIFICATION

Applying Advanced Functions & Equations principles, $g(-1) = 7$, then $f(7) = -27$. Therefore, the verified correct option is -27.

16 MCQ

If $f(x) = 2x + 7$ and $g(x) = 3x - 2$, find $f(g(-5))$.

A -18	**B -27** ✓
C -11	D -25

WORKED SOLUTION & VERIFICATION

Applying Advanced Functions & Equations principles, $g(-5) = -17$, then $f(-17) = -27$. Therefore, the verified correct option is -27.

17 MCQ

If $f(x) = -3x - 5$ and $g(x) = -4x$, find $f(g(-1))$.

A 2	**B -17** ✓
C 8	D -20

WORKED SOLUTION & VERIFICATION

Applying Advanced Functions & Equations principles, $g(-1) = 4$, then $f(4) = -17$. Therefore, the verified correct option is -17.

18 MCQ

If $f(x) = -5x - 3$ and $g(x) = -x - 3$, find $f(g(-4))$.

A -8 ✓	B 21
C -20	D -13

WORKED SOLUTION & VERIFICATION

Applying Advanced Functions & Equations principles, $g(-4) = 1$, then $f(1) = -8$. Therefore, the verified correct option is -8.

19 MCQ

If $f(x) = -2x - 2$ and $g(x) = 2x - 3$, find $f(g(-1))$.

A -2	**B 8** ✓
C -3	D 6

WORKED SOLUTION & VERIFICATION

Applying Advanced Functions & Equations principles, $g(-1) = -5$, then $f(-5) = 8$. Therefore, the verified correct option is 8.

20 MCQ

If $f(x) = 5x + 3$ and $g(x) = -x - 2$, find $f(g(-1))$.

A -1

B 0

C 3

D **-2** ✓

WORKED SOLUTION & VERIFICATION

Applying Advanced Functions & Equations principles, $g(-1) = -1$, then $f(-1) = -2$. Therefore, the verified correct option is -2.

21 MCQ

If $f(x) = -2x + 3$ and $g(x) = -4x + 3$, find $f(g(-2))$.

A -25

B -21

C **-19** ✓

D 15

WORKED SOLUTION & VERIFICATION

Applying Advanced Functions & Equations principles, $g(-2) = 11$, then $f(11) = -19$. Therefore, the verified correct option is -19.

22 MCQ

If $f(x) = x + 4$ and $g(x) = 4x + 6$, find $f(g(4))$.

A 27

B 24

C **26** ✓

D 38

WORKED SOLUTION & VERIFICATION

Applying Advanced Functions & Equations principles, $g(4) = 22$, then $f(22) = 26$. Therefore, the verified correct option is 26.

23 MCQ

If $f(x) = x - 3$ and $g(x) = -x + 2$, find $f(g(0))$.

A -3

B 5

C **-1** ✓

D 0

WORKED SOLUTION & VERIFICATION

Applying Advanced Functions & Equations principles, $g(0) = 2$, then $f(2) = -1$. Therefore, the verified correct option is -1.

24 MCQ

If $f(x) = -2x + 2$ and $g(x) = -4x - 5$, find $f(g(5))$.

A -28	B 27
C **52** ✓	D 50

WORKED SOLUTION & VERIFICATION

Applying Advanced Functions & Equations principles, $g(5) = -25$, then $f(-25) = 52$. Therefore, the verified correct option is 52.

25 MCQ

If $f(x) = 3x + 4$ and $g(x) = x - 5$, find $f(g(-1))$.

A 0	B **-14** ✓
C -4	D -11

WORKED SOLUTION & VERIFICATION

Applying Advanced Functions & Equations principles, $g(-1) = -6$, then $f(-6) = -14$. Therefore, the verified correct option is -14.